

Speech Understanding (CSL7770)

Assignment 2 Report

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The code can be found here: [GitHub Repository](#)

QUESTION 1 - SPEECH ENHANCEMENT

The task required us to use the VoxCeleb1 and VoxCeleb2 datasets to test a pre-trained model (UniSpeech SAT Large was used) and then fine-tune it. We had to report EER, TAR@1%FAR and Speaker Identification Accuracy.

UniSpeech SAT Large model was used for fine-tuning and evaluation. The results are reported in the following table 1.

Table 1. Results

	EER (%)	TAR@1%FAR	Identification Acc. (%)
Pre-trained	30.08	16.27	-
Finetuned	10.72	45.61	20.42

HuggingFace transformers [1] and Pytorch was used to achieve the task.

QUESTION 2 - MFCC Feature Extraction and Comparative Analysis of Indian Languages

TASK A

The task required us to extract Mel-Frequency Cepstral Coefficients (MFCC) from the given audio dataset containing samples of 10 different Indian languages. And also, needed to visualize MFCC spectrograms for at least three chosen languages, and compare the spectral patterns across these languages. Finally, perform a statistical analysis of the MFCC coefficients.

MFCCs were extracted using the mfcc feature extractor of librosa [2] Python package. And matplotlib [3] Python package was used for saving spectrograms and other language related statistics.

To analyze the **spectrograms**, 20 random spectrograms were saved for all the languages.

For **spectral analysis**, mean, standard deviation, minimum and maximum of all the MFCC features were plotted for all the languages. The plots of the first 4 MFCC statistics plots are attached. 1 2 3 4

The following analysis compares MFCC 0, 1, and 2 metrics — mean, standard deviation (Std), minimum, and maximum for all ten Indian languages.

MFCC 0, represents signal energy, depicts that Tamil has the highest mean and maximum values, which shows strong energy levels. Bengali and Malayalam have the lowest mean values. Telugu has the highest standard deviation, which shows high variability, while Kannada has the lowest standard deviation and maximum value, which implies consistence and lower energy.

MFCC 1, represents spectral shape, shows that Kannada has the highest mean and lowest standard deviation which shows stability. Hindi has the lowest mean and high variability. Also, Kannada and Gujarati have positive minimum values.

MFCC 2, represents finer spectral details, shows Kannada has the lowest mean, minimum, and maximum values suggesting a wide and negatively skewed distribution. Hindi shows the highest maximum value, and Telugu displays the most variability.

For conclusion, Tamil displays high energy and consistent speech peaks; Kannada shows low energy and stable spectral patterns; Hindi and Telugu shows greater variability in spectral features.

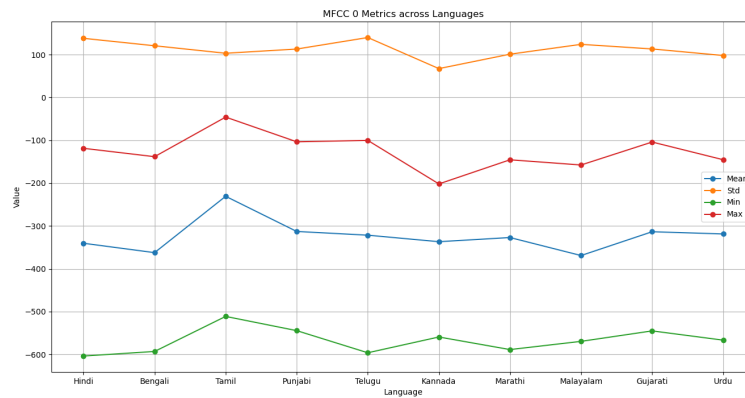


Fig. 1. MFCC 0

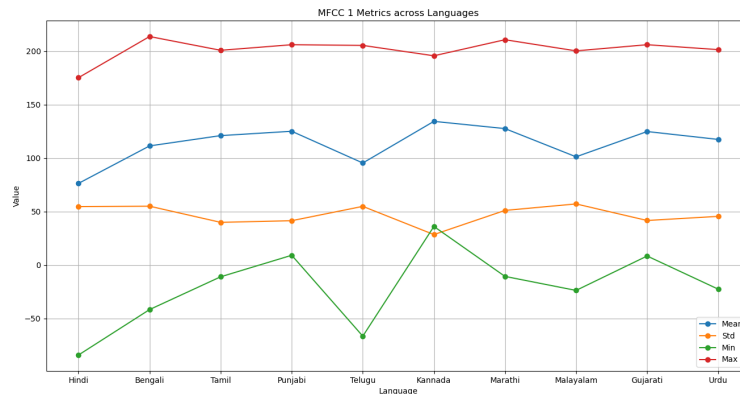


Fig. 2. MFCC 1

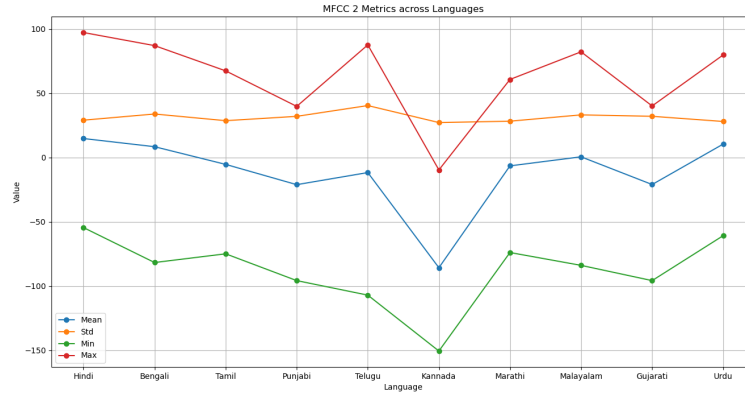


Fig. 3. MFCC 2

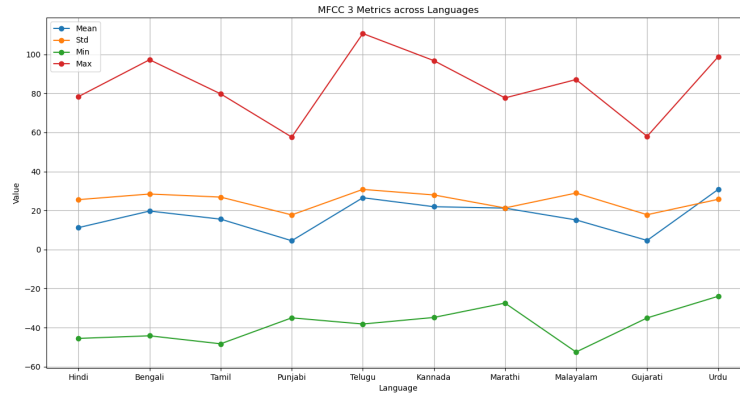


Fig. 4. MFCC 3

TASK B

The task required us to use the extracted MFCC spectral information to train models for language classification.

The features extracted from the above method were stored in one CSL file for each language. During training the data was combined into one dataset to be used.

Then, as instructed, 80:20 train-test split was applied to the dataset. and the features were standardized using a *Standard Scaler* to normalize their distributions as the models are sensitive to order of the features.

Then, five different classification models were trained and tested on the dataset. Grid Search was used for ML models (except SVM) and the optimal parameters received are as follows:

- K-Nearest Neighbors (KNN)
`n_neighbors = 9, weights = 'uniform'`
- Random Forest (RF)
`n_estimators = 50, max_depth = 20, min_samples_split = 2`
- Decision Tree Classifier (DTC)
`criterion = 'gini', max_depth = 20, min_samples_split = 2`
- Support Vector Classifier (SVC)
`C = 0.1, degree = 3, kernel = 'linear'`
- Neural Network (NN)
Layer Structure: 52-26-10

The accuracy and confusion matrix of each model was recorded in the respective log file. The accuracies for all the models are reported in Table 2.

Table 2. Training and Testing Accuracy of Different Models

Model	Train Accuracy (%)	Test Accuracy (%)
K-Nearest Neighbors (KNN)	89.94	84.16
Decision Tree (DTC)	87.91	80.76
Support Vector Machine (SVM)	81.73	81.47
Random Forest (RF)	91.05	81.55
Neural Network (NN)	87.31	87.29

All the above was done using the scikit-learn Python package.

References

1. https://huggingface.co/docs/transformers/model_doc/unispeech-sat.
2. <https://librosa.org/doc/latest/index.html>.
3. <https://matplotlib.org/>.